Module-2

Random Experiments:

* An exp that result in diff output eventhough it is repeated in a same manner everytime is called Random exp.
* The set of all posible outcomes of a random exp is called sample space of an exp.
* A sample space is discrete if it consists of a finite and countable infinite set of outcomes.
* A sample space is continuous if it contains an interval of real numbers.
* An event is a subset of the sample space of a random exp

1) Consider the random exp of tossing a fair coin twice. Write down the sample space of this exp. State whether the sample space is discrete or continuous and explain why? Define an event where atleast one head appears and list its outcomes.

ans: $S = \{HT, TH, TT, HH\}$

Discrete. Finite set of outcomes.

$E = \{HT, TH, HH\}$

2) If E_1 = 'first toss is head' i) find $E_1 \cup E_2$ and $E_1 \cap E_2$

E_2 = 'Second toss is head'.

$E_1 = \{HT, HH\}$ $E_1 \cup E_2 = \{HT, TH, HH\}$

$E_2 = \{TH, HH\}$ $E_1 \cap E_2 = \{HH\}$

3) Write the complement of Event E = "atleast one head appears"

$E = \{HT, TH, HH\}$ $E^c = \{TT\}$

4) Give an eg of 2 mutually exclusive event in this exp.

ans E_1 = Both tosses are head $\{HH\}$ $E_1 \cap E_2 = \phi$

E_2 = Both tosses are tail $\{TT\}$

* The union of 2 events is a event that consists of all outcomes that are contained in either of the 2 events. We denote it as $E_1 \cup E_2$.

* The intersection of 2 events E_1 and E_2 is a event that consists of all outcomes that are contained in both 2 events $E_1 \cap E_2$.

* Complement of an event E is a set of all outcomes in the sample space that are not in the event.

E' or E^c

* 2 events E_1 and E_2 are said to be mutually exclusive event if $E_1 \cap E_2 = \phi$

Probability of an Event

The probability of an event E is obtained by dividing the number of favourable outcomes by by total no. of outcomes.

$$\text{Prob of an event} = \frac{\text{No. of favourable outcome}}{\text{Total no. of outcomes}}$$

$$P(E) = \frac{n(E)}{n(S)}$$

Note: Whenever a sample space consists of 'N' pomible outcomes that are equally likely, the probability of each

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Note :- Whenever a sample space consists of 'N' pomible outcomes that are equally likely, the probability of ea outcome is $\frac{1}{N}$.

Axioms of Probability

Prob is a number that is assigned to each member of a collection of events from a random exp that satisfies the following properties * If 'S' is the sample space and 'E' is any event in the random exp, $P(S) = 1$

* $0 \leq P(E) \leq 1$
* For any 2 ~~mutually exclusive~~ events E_1 and E_2,
$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$
* If E_1 and E_2 are mutually exclusive,
$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(\phi)$$
$$= P(E_1) + P(E_2)$$
* Probability of an empty set is 0.
$$P(\phi) = 0$$
* For any event E, probability of its complement, $P(E') = 1 - P(E$
* For if the event E_1 is contained in the event E_2, then probabilty
$$P(E_1) \leq P(E_2)$$
* For any 3 events, E_1, E_2, E_3
$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) -$$
$$P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3)$$
* For a collection of mutually exclusive events $E_1, E_2, \ldots E_k$
$$P($$

1) Each of the 5 possible outcome of random exp is equally likely & $S = \{A, B, C, D, E\}$. Let $A = \{A, B\}$
$B = \{C, D, E\}$

Determine the following.

a) $P(A) = \frac{2}{5}$ $\quad 1 - P(A) \quad 1 - \frac{2}{5}$

b) $P(B) = \frac{3}{5}$

c) $P(A^c) = \frac{3}{5}$

d) $P(A \cup B) = \frac{2}{5} + \frac{3}{5} - 0 = 1$

e) $P(A \cap B) = 0$



2) If $P(A) = 0.3$, $P(B) = 0.2$, $P(A \cap B) = 0.1$ $\quad A^1 = 0.7$ $\quad P(B) = 0.2$

1) $P(A') = 1 - 0.3 = 0.7$

2) $P(A \cup B) = 0.3 + 0.2 - 0.1 = 0.4$

3) $P(A' \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = 0.1$

4) $P(A \cap B') = P(A) - P(A \cap B) = 0.3 - 0.1 = 0.2$

5) $P(A \cup B)' = 1 - P(A \cup B) = 1 - 0.4 = 0.6$

6) $P(A' \cup B) = 1 - P(A) + P(A \cap B)$
$= 1 - 0.3 + 0.1$
$= 0.8$



3) In a city 60% of all household get internet service from the local cable company, 80% get tv service from that company and 50% get both services from that company. If a household is randomly selected, i) what is the prob that it gets atleast one of these 2 services from that company.

iii) What is ~~prob~~ probability that it gets exactly one of the services for the company.

Let A be event of getting internet services.

B = get tv services

ans:

$$P(A) = \frac{60}{100} = 0.6 \quad P(B) = \frac{80}{100} = 0.8$$

$$P(A \cap B) = \frac{50}{100} = 0.5$$

i) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.8 - 0.5 = \underline{0.9}$

ii) Exactly $= P(A) + P(B) - 2P(A \cap B)$

$= 0.6 + 0.8 - 2 \times 0.5 = \underline{0.4}$

Conditional Probability

The probability of an event 'B' under the knowledge that the outcome will be in event 'A' is denoted as $P(B|A)$ and this is called the conditional probabilty of B given A.

Eg:- In a deck of 52 cards, the 2 cards are being drawn consider the following events.

A : Getting a red card on first draw.

B: Getting a red card on second draw

Find the conditional probabilty of getting a red card on the second draw that we got a red card on the 1st draw.

ans: $P(B|A) = \frac{25}{51}$ After drawing red card on the 1st draw there are 51 cards remaining and there are 25 red cards.

got $52 - 1 = 51$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Two events are independent any one of the following equ statements are true.

i) $P(A|B) = P(\overset{A}{B})$

ii) $P(A \cap B) = P(A) \cdot P(B)$

iii) $P(B|A) = P(B)$

Bayes Theorem

If $E_1, E_2, \ldots E_n$ are n mutually exclusive and exhaustive event and B is any event, then $P(E_i|B) = P(E_i) \cdot P(B|$

$$\frac{}{\sum_{k=1}^{n} P(E_k) \cdot P(B|}$$

Eg:- Bag 1 contains 4 white and 6 black balls and Bag 2 contains 4 white and 3 black balls. One ball is drawn from one of these bags and found to be black. Find the prob th it was drawn from Bag 1.

ans:

	White	Black
Bag 1	4	6
Bag 2	4	3

$\text{Prob}(\text{Bag 1}_{\text{Black}}) = \frac{6}{10}$

$\text{Prob}(\text{Bag 2}_{\text{black}}) = \frac{3}{7}$

E_1 = Choosing the Bag 1

E_2 = Choosing the Bag 2

B = Drawing a black ball.

$P(E_1) = \frac{1}{2}$

$P(E_2) = \frac{1}{2}$

$$P(E_1|B) = \frac{3/10 \times \frac{140}{76}}{\frac{91}{140}}$$

$$\frac{1/2}{\frac{3}{10} + \frac{3}{14}}$$

$$= \frac{1/2 \times 6/10}{1/2 \times 6/10 + 1/2 \times 3/7}$$

$$= 7/12$$

Random Variable

→ It is a function that assigns a real number to each outcome in the sample space of the random exp.

→ A discrete RV is a RV with finite range / countably infinite range.

Eg:- The no. of heads in 4 tosses of a coin is a discrete random variable as it can assume only values

Sample space = {HHHH, HTTT, ... TTTT, ...}.

X = no. of heads in 4 toss of a coin.

x can assume values → 0, 1, 2, 3, 4

TTTT, HTTT ... HHHH

→ A continuous Random variable is a random variable which can assume any value within an interval.

Eg:- The heights of a group of individuals.

→ Discrete Probability distribution.

Let RV 'x' assume values $x_1, x_2, x_3, \ldots, x_n$ with probabilities $P_1, P_2, P_3, \ldots P_n$ where $P(X = x_i) = P_i \geq 0$ and $P_1 + P_2 + \ldots + P_n = 1$ then,

X	x_1	x_2	x_3	...	x_n
$P(X=x_i)$	P_1	P_2	P_3	--	P_n

is called discrete probability distribution of X and it says how a total prob of 1 is distributed over several values of the random variable.

Continuous Probability distribution

A random variable is a continuous variable, if its prob distribution is called continuous probability density function (pdf). 'x' is a continuous R.V, then the pdf $f(x)$ sa the following properties.

1. $f(x) \geq 0$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$

Pdf can be used to determine the probabilities of the continuous RV.

i) $P(X \geq 0) = \int_{0}^{\infty} f(x)dx$

ii) $P(X \leq 0) = \int_{-\infty}^{0} f(x)dx$

ii) $P(a \leq x \leq b) = \int_{a}^{b} f(x)dx$

1) A RV 'x' has the following pdf

Value of x	-2	-1	0	1	2	3
P(x)	0.1	k	0.2	2k	0.3	k

i) Find value of k.

$$0.1 + k + 0.2 + 2k + 0.3 + k = 1$$

$$.4k + 0.6 = 1$$

$$.4k = 0.4$$

$$k = \frac{0.4}{4} = 0.1$$

2) Given that $f(x) = kx^4 e^{-x}$, $0 < x < \infty$ is a pdf. Determine value of k.

ans:
$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^{\infty} kx^4 e^{-x} = 1$$

$$k\int_0^{\infty} x^5 e^{-x} \quad k\int_0^{\infty} x^4 e^{-x} = 1$$

$$\int_0^{\infty} e^{-x} x^{n-1}\,dx = \Gamma(n) = (n-1)!$$

gamma

$$24k = 1$$

$$k = 1/24 \quad \frac{(3+1)!}{24}$$

3) 5 defective bulbs are accidently mixed with 20 good ones, is not possible to just look at a bulb and tell whether is defective. Find the prob distribn of no. of defective bulbs if 4 bulbs are drawn at random from this slot.

ans: $x = 20$

$x + 5 = 25$

Let 'x' be the no. of defective bulbs among the 4 bulbs drawn at random.

	G	D
25	20	5

x	0, 1, 2, 3, 4
P(x)	

$$P(X=0) = P(\text{all 4 are good ones}) = \frac{20C_4}{25C_4} = \frac{969}{25\,30}$$

$$P(X=1) = P(\text{all 3 are good ones}) + P(G=3)$$

$$= \frac{5C_1}{25C_1} * \frac{20C_3}{25C_3} = \frac{1140}{2530}$$

$$nC_p = \frac{n!}{(n-p)!}$$

$$\frac{20 \times 19 \times 18 \times 17}{1 \times 2 \times 3 \times 4}$$

$$P(X=2) = P(G=2) + P(D=2)$$

$$= \frac{20C_2}{25C_2} * \frac{5C_2}{25C_2} = \frac{20C_2 * 5C_2}{25C_4} = \frac{380}{2530}$$

$$P(X=3) = \frac{20C_1 * 5C_3}{25C_4} = \frac{1140}{2530}$$

$$P(X=4) = \frac{5C_4}{25C_4}$$

X	0	1	2	3	4
P(X)	$\frac{969}{2530}$	$\frac{1140}{2530}$	$\frac{380}{2530}$	$\frac{40}{2530}$	$\frac{1}{2530}$

Distribution function

'X' is a random variable $P(X \leq x)$ is called the distribution function and is denoted by $f(x)$.

$$F(x) = P(X \leq x) = \begin{cases} \sum_{y, y \leq x} P(X=y), & X \text{ is a discrete} \\ \int_{-\infty}^{x} f(x)\,dx, & X \text{ is a continuous} \end{cases}$$

1) A R.V 'X' has a following prob. function.

Value of X : x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

i) Find k

ii) Evaluate $P(X<6)$, $P(X\geq 6)$, $P(3<X\leq 6)$

ans: i) $0+k+2k+2k+3k+k^2+2k^2+7k^2+k=1$

$$9k+10k^2=1$$

$$10k^2+9k-1=0$$

$$-1 \text{ or } \frac{2}{20}=\frac{1}{10} \qquad \frac{-9\pm\sqrt{121}}{20} \qquad \frac{-9\pm\sqrt{81-4\times10\times-1}}{2\times10}$$

ii) $P(X<6)$

$= P(X=0,\ldots X=5)$

$= 0+0.1+2\times0.1+2\times0.1+3\times0.1+(0.1)^2$

$= 0.1+0.2+0.2+0.3+0.01$

$= 0.81 = \frac{81}{100}$

iii) $P(X\geq 6) = 2\times(0.1)^2+7(0.1)^2+0.1$

$= \frac{19}{100}$

iv) $P(3<X\leq 6) = P(X=4)+P(X=5)+P(X=6)$

$= (3\times0.1)+(0.1)^2+(2\times(0.1)^2)$

$= \frac{3}{10}+\frac{1}{100}+\frac{2}{100} = \frac{33}{100}$

Expected Values and Variance of a Random Variable

Let 'X' be a discrete random variable with pdf = $p(x)$, the expected value or mean value of 'X', denoted by $E(X)$ or μ_x or μ.

$$\boxed{E(X)=\sum_{x\in D} x\,p(x)}$$

D is set of possible values of X.

Expected / Mean value of continuous R.V 'X', with pdf $f(x)$ is, $\boxed{E(X) = \int_{-\infty}^{\infty} x f(x)\, dx}$.

Variance of a random variable 'X',

Let 'X' be a discrete R.V with pdf, $p(x)$ and expected value μ, then the variance of 'X', denoted by $V(X)$ or or σ_x^2 is, $V(X) = \sum_{x \in D} E(X-\mu)^2 = \sum_{x \in D} (x-\mu)^2 p($

Standard deviation of 'X' is, $SD = \sqrt{V(x)} = \sqrt{\sigma_x^2} = \sigma$

The variance of a continuous R.V 'X', with pdf, $f(x)$ and mean value μ ($\mu = E(X)$) is given by,

$$V(X) = E(X-\mu)^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx$$

Properties:

1) $E(k) = k$, k is a constant
2) $E(kX) = k E(X)$
3) $E(aX+b) = a E(X) + b$
4) $E(X^2) = \begin{cases} \sum x^2 p(x), & X \text{ is d.r.v} \\ \int x^2 f(x)\, dx, & X \text{ is c.r.v} \end{cases}$
5) $V(X) = E(X^2) - [E(X)]^2$
6) $V(k) = 0$, k is a constant.
7) $V(k(X) = k^2 V(X)$
8) $V(aX+b) = a^2 V(X)$

1) Let 'x' be a number on a die when thrown. Find the mean, variance, and SD of 'x'.

ans:

X = no. of numbers on a die when thrown

x	1	2	3	4	5	6
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\text{Mean} = E(X) = \sum x\,p(x)$$

$$= \frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6} = \frac{21}{6} = \frac{7}{2}$$

$$\text{Variance} = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum x^2 p(x)$$

$$= \frac{1}{6} + 2^2 \times \frac{1}{6} + 3^2 \times \frac{1}{6} + 4^2 \times \frac{1}{6} + 5^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6}$$

$$= \frac{1}{6} + \frac{4}{6} + \frac{9}{6} + \frac{16}{6} + \frac{25}{6} + \frac{36}{6}$$

$$= \frac{91}{6}$$

$$V(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

$$S.D = \sqrt{V(X)} = \sqrt{\frac{35}{12}}$$

2) Mean and variance of a r.v with pdf, $f(x) = k(1-x^2), 0<x$ find the

ans: $f(x) = k(1-x^2)$ pdf: $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_0^1 k(1-x^2)\,dx = 1$$

$$k\left[\left(1 - \frac{1}{3}\right)\right] = 1$$

$$k\int_0^1 (1-x^2)\,dx = 1$$

$$k \times \frac{2}{3} = 1$$

$$k\left[x - \frac{x^3}{3}\right]_0^1 = 1$$

$$2k = 3$$

$$k = \frac{3}{2}$$

$$f(x) = \frac{3}{2}(1-x^2),\ 0<x<1$$

$$E(x) = \int_0^1 x f(x)\,dx = \int_0^1 x \cdot \frac{3}{2}(1-x^2)\,dx$$

$$= \frac{3}{2}\int_0^1 x(1-x^2)\,dx$$

$$= \frac{3}{2}\int_0^1 (x-x^3)\,dx = \frac{3}{2}\left[\frac{x^2}{2} - \frac{x^4}{4}\right]$$

$$= \frac{3}{2}\left[\frac{1}{2} - \frac{1}{4}\right] = \frac{3}{2}\left[\frac{1}{4}\right] = \frac{3}{8}$$

$$V(X) = E(X^2) - E(X)^2$$

$$E(X^2) = \int_0^1 x^2 f(x)\,dx = \int_0^1 x^2 \cdot \frac{3}{2}(1-x^2)$$

$$= \frac{3}{2}\int_0^1 x^2(1-x^2) = \frac{3}{2}\int_0^1 (x^2 - x^4)\,dx$$

$$= \frac{3}{2}\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \frac{3}{2}\left[\frac{1}{3} - \frac{1}{5}\right] = \frac{3}{2}\left[\frac{2}{15}\right]$$

$$= \frac{6}{30} = \frac{1}{5}$$

$$V(X) = \frac{1}{5} - \left(\frac{3}{8}\right)^2$$

$$= \frac{1}{5} - \frac{9}{64} = \frac{19}{320}$$

1) Find the expected number of throws of an unbiased die till face marks 5 turns up.

ans: X : no. of throws till the face '5' turns up.

X	1	2	3	4	...
$P(X)$	$\frac{1}{6}$	$\frac{5}{6}\times\frac{1}{6}$	$\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6}$	$\frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6}$	

$P(X=2)$ = P (getting a num $\neq 5$ in 1st toss & getting 5 in next)

$= \frac{5}{6}\times\frac{1}{6}$

$$P(X=n) = \left(\frac{5}{6}\right)^{n-1}\left(\frac{1}{6}\right)$$

$$E(X) = \sum_{n=1}^{\infty} n\,p(n)$$

$$= 1\cdot\left(\frac{1}{6}\right) + 2\cdot\left(\frac{5}{6}\right)\left(\frac{1}{6}\right) + \cdots$$

$S_\infty = \frac{a}{1-r}$

1) Obtain the distribution function and mean of the total no. of heads occuring in 3 tosses of an unbiased coin

ans: X : no. of heads obtained in 3 tosses. 0, 1, 2, 3

X : 0 1 2 3

$P(X=0) = \frac{1}{8}$ $P(X=2) = \frac{3}{8}$

$P(X=1) = \frac{3}{8}$ $P(X=3) = \frac{1}{8}$

Mean, $E(X) = \sum n\,p(n)$

$= 0 + \frac{3}{8} + \frac{6}{8} + \frac{3}{8} = \frac{12}{8}$

2) A die is tossed thrice, a success is getting '1' or '6' on a toss. Find the mean of the no. of successes.

ans: X : no. of success $p(X=0) = P[\text{all are failures}] = \left(\frac{4}{6}\right)^3$

X : 0 1 2 3 $= \left(\frac{2}{3}\right)^3$

$P[\text{success}] = \frac{2}{6} = \frac{1}{3}$

$P[\text{failure}] = \frac{4}{6} = \frac{2}{3}$

$P(X=1) = P[1 \text{ success and } 2 \text{ failure}] = 3\left[\frac{1}{3} \times \frac{2}{3} \times \frac{2}{3}\right]$

$P(SFF) = \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3}$ $= \frac{12}{27}$

$P(FSF) = \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3}$

$P(FFS) = \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3}$

$P(X=2) = P[2 \text{ success and } 1 \text{ failure}] = 3\left[\frac{1}{3} \times \frac{1}{3} \times \frac{2}{3}\right]$

$= \frac{6}{27}$

$P(X=3) = P[3 \text{ success}] = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$.

$\text{Mean} = \sum x\,p(x) = 0 \times \frac{8}{27} + 1 \times \frac{12}{27} + 2 \times \frac{6}{27} + 3 \times \frac{1}{27}$

$= 0 + \frac{12}{27} + \frac{12}{27} + \frac{3}{27} = \frac{27}{27} = 1$.

Jointly Distributed Random Variable

The pdf of a single discrete R.V 'x' specifies how much probability mass is placed on each possible value of 'x', whereas the joint pdf of 2 discrete R.V 'x' and 'y' describes how much probability mass is placed on each possible pair of values (x, y).

→ Let 'x' and 'y' be 2 discrete R.V defined on the sample space 'S' of an experiment. The joint probability mass function $p(x, y)$ is defined for each pair of (x, y)

$p(x, y) = P(X=x, Y=y)$

- where, $p(x,y) \geq 0$, $\sum_x \sum_y p(x,y) = 1$

→ Let 'X' and 'Y' be continuous R.V, a joint pdf $f(x,y)$ for these 2 random variables is a function satisfy

$f(x,y) \geq 0$, $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dy\, dx = 1$

→ Then for any 2 dimensional set $\{A\}$, probability that

$P[(X,Y) \in A] = \iint_A f(x,y)\, dx\, dy$

In particular if A is the 2D rectangle, $\{(x,y) \mid a \leq x \leq b, c \leq y \leq d\}$ then $P[(X,Y) \in A] = P(a \leq x \leq b, c \leq y \leq d)$

$$= \int_{y=c}^{d} \int_{x=a}^{b} f(x,y)\, dx\, dy$$

1) Show that the following function satisfies the properties of a joint probability mass function.

x	y	f(x,y)
1	1	1/4
1.5	2	1/8
1.5	3	1/4
2.5	4	1/4
3	5	1/8

a) $P(X < 2.5, Y < 3)$

b) $P(X < 2.5)$

c) $P(Y < 3)$

d) $P(X > 1.8, Y > 4.7)$

ans:

$f_{xy}(x,y) \geq 0$, $\forall (x,y)$

$\sum_x \sum_y f_{xy}(x,y) = \frac{1}{4} + \frac{1}{8} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} = 1$

∴ This is a valid distributed function.

$$P[X<2.5, Y<3] = P[X=1, Y=1] + P[X=1.5, Y=2]$$

$$= \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$P[X<2.5] = P[X=1, Y=1] + P[X=1.5$$

$$= P[1] + P[1.5] + P[1.5] = \frac{1}{4} + \frac{1}{8} + \frac{1}{4} = \frac{5}{8}$$

$$P[Y<3] = P[Y=1] + P[Y=2] = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$P[X>1.5, Y>4.7] = P[X=3, Y=5] = \frac{1}{8}$$

2) A bank operates both a drive-up facility and a walk-up facility on a randomly selected day. Let 'X' be the proportion of time that the drive-up facility in use. (atle one customer is being served or waiting to be served and be the proportion of time), 'Y' is the proportion of time walk-up fac Then set (X, Y) is a rectangle $D = \{(x, y) \mid 0 \le x \le 1, 0 \le y \le$

Suppose $f(x,y) = \begin{cases} \frac{6}{5}(x+y^2), & 0 \le x, y \le 1 \\ 0, & \text{otherwise.} \end{cases}$

Verify whether $f(x,y)$ can serve as a valid joint pdf for $(x$ If it is valid, determine the prob that neither facility is bu more than one quarter of the time

Soln: a) $f(x,y) \ge 0$

$$\iint_D f(x,y)\,dx\,dy = \int_{y=0}^{} \int_{x=0}^{} \frac{6}{5}(x+y^2)\,dx\,dy$$

Ans:

a) $\frac{6}{5}\int_{y=0}^{1}\int_{x=0}^{1}(x+y^2)\,dx\,dy$

$\Rightarrow \frac{6}{5}\int_{y=0}^{1}\left[\frac{x^2}{2}\right]_0^1+y^2\,dy$

$\Rightarrow \frac{6}{5}\int_0^1 \frac{1}{2}+y^2\,dy \Rightarrow \frac{6}{5}\times\frac{1}{2}\left[\frac{y^3}{3}\right]_0^1$

$\Rightarrow \frac{6}{5}\int_0^1\left(\frac{1}{2}+y^2\right)dy \Rightarrow \frac{6}{5}\times\frac{1}{2}\times\frac{1}{3}$

$\Rightarrow \frac{6}{5}\left[\left(\frac{1}{2}\right)y+\frac{y^3}{3}\right]_0^1 \Rightarrow \frac{6}{5}\left[\frac{1}{2}+\frac{1}{3}\right]$

$\Rightarrow \frac{6}{5}\left[\frac{5}{6}\right]=1$

$\therefore$ This is valid pdf.

b) $P\left[0\le X\le\frac{1}{4},\ 0\le Y\le\frac{1}{4}\right]$

$\Rightarrow \int_0^{1/4}\int_0^{1/4}\frac{6}{5}(x+y^2)\,dx\,dy \Rightarrow \frac{6}{5}\int_0^{1/4}\left[\frac{x^2}{2}+y^2x\right]_0^{1/4}dy$

$\Rightarrow \frac{6}{5}\int_0^{1/4}\left[\frac{(\frac{1}{4})^2}{2}+y^2\frac{1}{4}\right]dy \Rightarrow \frac{6}{5}\int_0^{1/4}\left[\frac{1}{32}+y^2\times\frac{1}{4}\right]dy$

$\Rightarrow \frac{6}{5}\left[\frac{1}{32}+\frac{1}{4}\left[\frac{y^3}{3}\right]\right]_0^{1/4} \Rightarrow \frac{6}{5}\left[\frac{1}{8}+\frac{1}{4}\times\frac{64}{3}\right]$

$\Rightarrow \frac{6}{5}\left[\frac{1}{8}+\frac{1}{4}\times\frac{1}{64}\times 3\right]$

$\Rightarrow \frac{1}{8}+\frac{1}{4}\times\frac{3}{64}$

$\Rightarrow \frac{6}{5}\left[\frac{y}{32}+\frac{y^2}{12}\right]_0^{1/4} \Rightarrow \frac{6}{5}\left[\frac{1}{128}+\frac{1}{768}\right] \Rightarrow \frac{7}{640}$

$\frac{3}{256}$

Marginal Probability Distribution

In a random exp involving more than 1 random variable, the individual probability distribution of a R.V is referred to as its marginal probability distribution and it can be determined by the Joint Probability distribution of the R.V.

Note 1 :-

Consider 2 discrete random variable X and Y. The M.P. of X and Y are :-

$$f_X(x) = \sum_y P[X=x, Y=y]$$

$$f_Y(y) = \sum_x P[X=x, Y=y]$$

Note 2 :-

If X and Y are 2 continuous R.V with joint probability distribution, ~~then~~ $f_{XY}(x,y)$, then the marginal pdf

$$f_X(x) = \int f_{XY}(x,y)\, dy$$

$$f_Y(y) = \int_x^y f_{XY}(x,y)\, dx$$

where the 1st integral is over all points in range (x,y) for which $X=x$ and the 2nd integral for which $Y=y$

The conditional pdf of Y given $X=x$, $Y/X=x$ is

$$f_{Y/X}(y) = \frac{f_{XY}(x,y)}{f_X(x)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

1) The joint probability distribution of 2 d.r.v x and y is given,

y \ x	1	2	3
4	0.15	0.1	0.05
3	0.02	0.1	0.05
2	0.02	0.03	0.2
1	0.01	0.02	0.25

i) find the marginal dist of x and y also find prob
1) $P[Y=1|X=3]$
2) $P[Y=2|X=3]$

ans: Marginal, $f_x(x) = \sum_y P(x, Y=y)$

$$f_x(1) = \sum_y P(1, Y=y)$$
$$= P(1,4) + P(1,3) + P(1,2) + P(1,1)$$
$$= 0.15 + 0.02 + 0.02 + 0.01 = 0.2$$

$$f_x(2) = \sum_y P(2, Y=y) = P(2,4) + P(2,3) + P(2,2) + P(2,1)$$
$$= 0.25$$

$$f_x(3) = \sum_y P(3, Y=y) = 0.55$$

$f_y(4) = 0.3$ $f_y(3) = 0.17$ $f_y(2) = 0.25$ $f_y(1) = 0.28$

Marginal pdf of $X = 0.2 + 0.25 + 0.55 = 1$

Marginal pdf of $Y = 0.3 + 0.17 + 0.25 + 0.28 = 1$

1) $P[Y=1|X=3] \Rightarrow \frac{f_{XY}(X=3, Y=1)}{f_x(3)} = \frac{P(X=3, Y=1)}{f(3)}$

$$= 0.25$$

b) $P[Y=2|X=3] \Rightarrow \frac{f_{xy}(x=3, y=2)}{f(3)} = \frac{0.2}{0.55}$

1) Let X and Y be 2 R.V and $f(x,y) = 6\times10^{-6} \cdot e^{-0.001x-0}$, $0<x<y$

a) Verify whether $f(x,y)$ can serve as a valid jpdf for (X,Y)

b) if it is valid, find the probability that $X<1000$ and $Y<200$

c) find the marginal pdf of Y. and hence calculate the prob that $Y>2000$

$$\int_0^\infty e^{-ax}\,dx = \frac{1}{a}$$

ans:

$$\int\int f(x,y)\,dx\,dy = 1$$

$$\int_0^\infty \int_0^y 6\times10^{-6} \cdot e^{-0.001x-0.002y}\,dx\,dy$$

$$6\times10^{-6}\int_0^\infty e^{-0.002y}\left[\frac{e^{-0.001x}}{-0.001}\right]_0^y dy$$

$$0.006\int_0^\infty \left(e^{-0.002y} - e^{-0.003y}\right.$$

* Explain Bayesian estimation of parameters.

→ Bayesian estimation is a statistical method that helps someone deal with conditional probability. It is done by using prior evidence to estimate an unknown population parameter.

Bayesian estimation is found by comparing the probability of event A and the prob of event B. The likelihood of these events occuring is compared to past evidence of these events to determine the likelihood of them occuring in the future.

Estimating unknown parameter using:

1. Prior knowledge about the parameter
2. Observed data
3. Bayes theorem

→ Suppose we want to estimate an unknown parameter θ.

Bayesian estimation uses:

$$P(\theta|X) = \frac{P(X|\theta)\,P(\theta)}{P(X)}$$

where, $P(\theta)$ = Prior distribution - what we believe about θ before seeing data.

$P(X|\theta)$ = Likelihood - how likely the observed data is for a particular θ.

$P(\theta|X)$ = Posterior distribution - Our updated belief about θ after seeing data. $P(\theta|X) \propto P(X|\theta) \cdot P(\theta)$

$P(X)$ = Evidence / normalizing constant. $P(X) = \int P(X|\theta)\,P(\theta)\,d$

So, Posterior $\propto$ Likelihood $\times$ Prior

Eg:- Estimating Probability of Heads,

Suppose we have a coin whose prob of getting hea
is unknown.

Let, $\theta = P(\text{head})$

1: Prior – Before tossing the coin, if we are believing th
all values of θ between 0 and 1 are equally li

This is represented by, $P(\theta) = 1, \quad 0 \leq \theta \leq 1$

This is a uniform prior

2: Observe data,

We toss a coin 10 times: Results are: H,H,T,H,
Suppose we get: Heads = 7 H,T,H
Tails = 3

$n = 10, \quad \alpha = 7.$

$n \rightarrow$ total no. of tosses
$\alpha \rightarrow$ no. of heads.

3: Find likelihood,

For one toss: $P(H) = \theta$

$\therefore P(T) = 1 - \theta$ because the coin must

produce either heads / Tails.

Prob of a particular sequence, Suppose probability is,

$$\theta \times \theta \times \theta \times \theta \times \theta \times \theta \times \theta \times (1-\theta) \times (1-\theta) \times (1-\theta)$$

$$\therefore = \theta^7 (1-\theta)^3$$

$$L(\theta) = \theta^7 (1-\theta)^3$$

There are many ways to get the arrangements, $\binom{10}{7} \Rightarrow {}^{10}C_7$

$${}^{10}C_7 = \frac{10!}{7!\,3!} = 120.$$

$\therefore$ Likelihood is : $\rightarrow$ becomes constant.

$$L(\theta) = {}^{10}C_7\, \theta^7 (1-\theta)^3$$

$$\therefore \quad L(\theta) \propto \theta^7 (1-\theta)^3$$

Apply Bayes theorem, $\underbrace{P(\theta|x)}_{\text{Posterior}} = \frac{\overbrace{P(x|\theta)}^{\text{likelihood}} P(\theta) \rightarrow \text{prior}}{\underbrace{P(x)}_{\text{evidence/normalizing constant}}}$

$$P(\theta|x) \propto P(x|\theta) P(\theta)$$

We found, $P(x|\theta) \propto \theta^7 (1-\theta)^3$

prior is, $P(\theta) = 1 \quad \therefore P(\theta|x) \propto \theta^7 (1-\theta)^3 \times 1$

hence, $P(\theta|x) \propto \theta^7 (1-\theta)^3 \Rightarrow$ This is the posterior distribution

A: Beta distribution, Beta (α, β)

whose density is proportional to: $\theta^{\alpha-1} (1-\theta)^{\beta-1}$

Our posterior is : $\theta^7 (1-\theta)^3$ from this

$$\alpha - 1 = 7 \quad \therefore \quad \alpha = 8.$$

$$\beta - 1 = 3 \quad \therefore \quad \beta = 4$$

$$\therefore \theta|x \sim \text{Beta}(8, 4)$$

$$\text{Mean} = E(\theta) = \frac{\alpha}{\alpha+\beta} = \frac{8}{8+4} = \frac{8}{12} = \frac{2}{3}$$

$$\therefore \hat{\theta}_{Bagged} = \frac{2}{3} \approx \underline{0.667}$$